Thrombocytopenia 9 (THC9): A Comprehensive Disease Characterization Report

Disease: Thrombocytopenia 9 (THC9) **MONDO ID:** MONDO:0957572 **Causal gene: THPO** (thrombopoietin; HGNC:11795; NCBI Gene 7066; UniProt P40225; chromosome 3q27.1) **Synonym:** THC9 **Category:** Mendelian, autosomal recessive

Summary

Thrombocytopenia 9 (THC9) is an ultra-rare inherited thrombocytopenia caused by biallelic loss-of-function variants in *THPO*, the gene encoding thrombopoietin (TPO). TPO is the master cytokine of platelet production: secreted constitutively by the liver, it binds its receptor MPL (c-Mpl) on megakaryocytes and hematopoietic stem cells (HSCs), triggering JAK2 activation and the downstream STAT5, PI3K–AKT, and MAPK/ERK signaling cascades that drive megakaryopoiesis and maintain the HSC pool. When TPO ligand activity is deficient, megakaryocyte and platelet production fails, producing a hypoproductive (central) thrombocytopenia. Critically, because the lesion lies in the ligand and **not** in the MPL receptor, THC9 is mechanistically distinct from receptor-defect disorders and is predicted to respond to receptor-level pharmacology.

The founding description of the disease (Dasouki et al., 2013, PMID: 24085763) identified a homozygous *THPO* missense variant, c.112C>T (p.Arg38Cys, also written R17C when the 21-amino-acid signal peptide is excluded), in the receptor-binding domain, in two siblings from a consanguineous Micronesian family with autosomal recessive aplastic anemia. The variant is a **partial** loss-of-function allele, and the family displayed a striking gene-dosage phenotypic gradient: homozygotes developed trilineage marrow failure (aplastic anemia), heterozygous carriers had mild thrombocytopenia, and a wild-type sibling had normal platelet counts. This dosage sensitivity places THC9 on a continuum from mild isolated thrombocytopenia to full marrow aplasia.

THC9 is best understood in contrast to its molecular mirror image. Whereas biallelic coding loss-of-function *THPO* variants reduce TPO activity and cause THC9/aplastic anemia, dominant gain-of-function mutations in the *THPO* 5′-untranslated region (5′UTR) increase TPO translation and instead cause hereditary (familial) essential thrombocythemia (THCYT1, OMIM 187950). The *THPO* gene therefore functions as a bidirectional rheostat that sets total body platelet mass. Because the MPL receptor remains intact in THC9, TPO-receptor agonists (romiplostim, eltrombopag) are a mechanistically rational therapy, with allogeneic hematopoietic stem cell transplantation (HSCT), and experimentally gene therapy, addressing the marrow-failure end of the spectrum.

---

1. Disease Information

THC9 is a Mendelian, autosomal-recessive inherited thrombocytopenia / bone marrow failure disorder caused by deficiency of thrombopoietin (TPO), the principal physiological regulator of platelet production. The clinical presentation ranges from isolated thrombocytopenia to aplastic anemia depending on allele dosage and residual TPO activity.

Key identifiers:

Resource	Identifier
MONDO	MONDO:0957572
Gene	*THPO* (HGNC:11795; NCBI Gene 7066)
OMIM (disease)	Not separately recorded in template; *THPO* gene = OMIM 600044
Synonym	THC9
Contrast disorder	THCYT1 / hereditary essential thrombocythemia, OMIM 187950 (5′UTR gain-of-function)

Synonyms / alternative names: Thrombocytopenia 9; THC9; (in the index family presented as) autosomal recessive aplastic anemia due to thrombopoietin deficiency.

Information source: The disease-level characterization is derived primarily from a small number of **individual patient reports** (notably a single consanguineous Micronesian family, Dasouki 2013) supplemented by **aggregated disease-level resources** (OMIM, MONDO) and extensive **model-organism and mechanistic literature** on the TPO–MPL axis. The evidence

base for THC9 as a distinct human entity is narrow (essentially one index family), so much of the phenotype and treatment rationale is extrapolated from TPO/MPL biology and from the related receptor disorder congenital amegakaryocytic thrombocytopenia (CAMT, MPL-related).

---

2. Etiology

Primary cause — genetic: Biallelic (homozygous or compound heterozygous) loss-of-function variants in *THPO*. In the index family the variant was homozygous c.112C>T (p.Arg38Cys / R17C), a missense change in the TPO receptor-binding domain. In vitro, the mutant TPO retained only a 2–3-fold reduced capacity to sustain TPO-dependent UT7-TPO cells, consistent with a **partial** loss-of-function allele rather than a null.

> "Whole-exome sequencing identified a novel homozygous missense mutation in thrombopoietin (THPO, c.112C>T) in both affected siblings." — PMID: 24085763
>
> "the R17C partial loss-of-function allele results in aplastic anemia in the homozygous state and mild thrombocytopenia in the heterozygous state in our family" — PMID: 24085763

Genetic risk factors: The causal variant itself is the risk factor; consanguinity is a strong contributor given the autosomal-recessive inheritance (the index family was consanguineous and Micronesian). Heterozygous carriers have a subclinical/mild thrombocytopenia risk (gene-dosage effect).

Environmental risk factors: None established. THC9 is a monogenic disorder; no toxins, lifestyle, or occupational exposures are known to cause or trigger it.

Protective factors: None specifically documented. By mechanistic inference, any residual TPO activity (as with the partial-LoF R17C allele retaining ~30–50% function) is protective against the most severe marrow-failure phenotype; a single wild-type *THPO* allele is sufficient for near-normal (mildly reduced) platelet counts.

Gene–environment interactions: No specific GxE interactions are documented for THC9. The monogenic, highly penetrant nature of the biallelic genotype leaves little room for environmental modification, though this is untested.

---

3. Phenotypes

The core phenotype is **hypoproductive (central) thrombocytopenia** with a dosage-dependent spectrum extending to pancytopenia/aplastic anemia in biallelic cases.

Phenotype	Type	HPO term	Onset / severity / frequency
Thrombocytopenia	Laboratory abnormality	HP:0001873	Carriers: mild; biallelic: severe. Core feature (~100% of affected)
Aplastic anemia / pancytopenia	Lab + clinical	HP:0001915 (aplastic anemia); HP:0001876 (pancytopenia)	Biallelic homozygotes; childhood-onset in index family
Anemia	Laboratory abnormality	HP:0001903	Part of biallelic marrow failure
Leukopenia / neutropenia	Laboratory abnormality	HP:0001882 / HP:0001875	Part of biallelic marrow failure
Abnormal bleeding / bruising	Clinical sign	HP:0001892 (abnormal bleeding)	Proportional to platelet count; mild in carriers
Reduced megakaryocytes	Histopathology	HP:0005528 (bone marrow hypocellularity)	Hypoproductive marrow

Phenotype characteristics: - **Age of onset:** Childhood in biallelic homozygotes (aplastic anemia in the index siblings); carriers show lifelong mild thrombocytopenia. - **Severity:** Variable and dosage-dependent — mild (heterozygotes) to severe (biallelic marrow failure). - **Progression:** The biallelic form is progressive toward marrow failure, consistent with the role of TPO/MPL in HSC maintenance (progressive HSC loss, as seen with MPL-inactivating mutations). Carrier thrombocytopenia appears stable. - **Frequency among affected individuals:** Thrombocytopenia is universal; the extension to trilineage aplasia depends on genotype and residual TPO activity.

Quality-of-life impact: Dominated by bleeding risk (in severe thrombocytopenia) and, in the biallelic marrow-failure form, by anemia-related fatigue, infection risk from neutropenia, and the burden of transfusion dependence or HSCT. Formal EQ-5D/SF-36 data specific to THC9 are **not available** given the rarity.

4. Genetic / Molecular Information

Causal gene: *THPO* (thrombopoietin), chromosome 3q27.1, HGNC:11795, NCBI Gene 7066, gene OMIM 600044. The protein TPO (UniProt P40225) comprises a 21-amino-acid signal peptide, an N-terminal receptor-binding (EPO-like) domain, and a heavily glycosylated C-terminal domain that stabilizes the circulating protein.

Pathogenic variants: - **Founding variant:** c.112C>T, p.Arg38Cys (p.Arg17Cys / R17C using mature-protein numbering that excludes the signal peptide). Located in the receptor-binding domain. - **Variant classification:** Pathogenic/likely pathogenic in the recessive, biallelic context, supported by functional data (2–3-fold reduced ability to sustain TPO-dependent cells) and co-segregation with a gene-dosage phenotype. - **Variant type:** Missense (partial loss-of-function). Other predicted loss-of-function classes (nonsense, frameshift, splice) would be expected to cause THC9 if biallelic, but the human evidence base is currently limited to the R17C missense allele. - **Allele frequency:** The index variant is novel/private (consanguineous family); THC9 alleles are vanishingly rare in gnomAD. - **Somatic vs germline:** Germline. - **Functional consequence:** Loss of function (reduced receptor-binding/signaling capacity of the TPO ligand). This is distinct from, and directionally opposite to, the gain-of-function 5′UTR mutations that cause hereditary thrombocythemia.

THPO directional genotype–phenotype dichotomy:

> "we identified a one-base deletion in the 5'-untranslated region of the TPO gene in affected but not in nonaffected family members. In vitro experiments showed that the identified mutation increased TPO production." — PMID: 9694695
>
> "The presence of AUG codons located upstream of the physiological start site can inhibit translation by causing premature initiation and thereby preventing the ribosome from reaching the physiological start codon, as in the case of thrombopoietin (TPO)." — PMID: 10828006

THPO variant class	Location	Effect on TPO	Phenotype	Inheritance
Missense / predicted LoF	Coding / receptor-binding domain	↓ TPO activity	**THC9** (thrombocytopenia → aplastic anemia)	Autosomal recessive
5′UTR deletion / uAUG disruption	5′UTR	↑ TPO translation/ production	Hereditary essential thrombocythemia (THCYT1)	Autosomal dominant

Modifier genes: Not specifically defined for THC9. Mechanistically, variants in *MPL*, *JAK2*, and downstream signaling genes would be expected to modify TPO responsiveness, but no modifiers are established.

Epigenetic information: *THPO* expression is partly controlled at the translational level by the 5′UTR uORF/stem-loop architecture rather than by classical DNA methylation/histone changes relevant to THC9. No disease-specific epigenetic signature is reported.

Chromosomal abnormalities: None characteristic; THC9 is a single-gene disorder, not a copy-number/structural syndrome.

---

5. Environmental Information

- **Environmental factors:** None causal. THC9 is monogenic.
- **Lifestyle factors:** None established.
- **Infectious agents:** Not applicable — THC9 is not infectious. (Note: TPO levels rise non-specifically in sepsis/inflammation, PMID: 35665097, which is relevant to diagnostic interpretation but not to THC9 causation.)

---

6. Mechanism / Pathophysiology

Ordered causal chain

1. A biallelic loss-of-function variant in *THPO* (e.g., c.112C>T / R17C in the receptor-binding domain) **results in** thrombopoietin protein with reduced receptor-binding/signaling capacity.
2. Reduced functional TPO **leads to** diminished engagement of the MPL (c-Mpl) receptor on megakaryocytes and hematopoietic stem cells.
3. Deficient MPL engagement **results in** reduced JAK2 activation and receptor tyrosine phosphorylation.
4. Reduced JAK2 signaling **leads to** attenuated downstream STAT5, PI3K–AKT, and MAPK/ERK cascade activity.
5. Attenuated signaling **results in** (branch A) impaired megakaryocyte proliferation, differentiation, and maturation → reduced platelet production → **thrombocytopenia**; and (branch B) impaired HSC maintenance/quiescence → progressive HSC attrition → **trilineage marrow failure** / **aplastic anemia** in biallelic cases.
6. Reduced anti-apoptotic (AKT–Bcl-xL) signaling in megakaryocytes/platelets **contributes to** decreased platelet survival, compounding the thrombocytopenia (inferred from TPO-RA apoptosis studies).
7. Thrombocytopenia **leads to** the clinical bleeding phenotype (bruising, mucocutaneous bleeding), proportional to platelet count.

Steps 1–5 are well supported by human genetics plus model-organism and in vitro data; the branch to full marrow failure (step 5B) is supported by MPL-null biology and the index family's aplastic anemia phenotype. The anti-apoptotic contribution (step 6) is inferred from TPO-RA pharmacodynamics rather than demonstrated directly in THC9 patients.

Supporting detail

Molecular pathways: The TPO→MPL→JAK2 axis activating STAT5, PI3K-AKT, and MAPK/ERK (KEGG hsa04630 JAK-STAT signaling; Reactome signaling by MPL).

> "Tpo binding to Mpl triggers activation of Janus kinase 2 (Jak2) and phosphorylation of the receptor, as well as activation of several intracellular signalling cascades that mediate cellular responses" — PMID: 38491305

Cellular processes: Megakaryocyte proliferation/differentiation/maturation (GO:0030219 megakaryocyte differentiation; GO:0030220 platelet formation), HSC maintenance and quiescence (GO:0061484 hematopoietic stem cell homeostasis), and anti-apoptotic regulation.

> "mutation of which resulted in thrombocytopenia, deficient megakaryopoiesis, low hematopoietic stem cell (HSC) number and function" — PMID: 38491305
>
> "Patients with inactivating mutations in Mpl develop thrombocytopenia and aplastic anemia because of progressive loss of HSCs" — PMID: 21289307

Protein dysfunction: Loss of function of the secreted TPO ligand — reduced affinity/efficacy at the receptor-binding interface (R17C) rather than misfolding or aggregation.

Anti-apoptotic / survival signaling: TPO-R signaling exerts anti-apoptotic effects on platelets and megakaryocytes via AKT-regulated Bcl-xL/Bax/Bak balance; loss of this input reduces platelet survival.

Immune system involvement: THC9 is a production defect, **not** an autoimmune or immune-destructive thrombocytopenia — an important distinction from immune thrombocytopenia (ITP). Neutropenia in the biallelic marrow-failure form predisposes to infection as a downstream complication.

GO / CL / UBERON / CHEBI term suggestions: GO:0038162 (thrombopoietin-mediated signaling pathway), GO:0030219 (megakaryocyte differentiation), GO:0030220 (platelet formation); CL:0000556 (megakaryocyte), CL:0000037 (hematopoietic stem cell), CL:0000233 (platelet); UBERON:0002371 (bone marrow), UBERON:0002107 (liver, site of TPO synthesis); CHEBI:51354 (eltrombopag).

---

7. Anatomical Structures Affected

Organ level: - **Primary:** Bone marrow (UBERON:0002371) — the site of defective megakaryopoiesis and HSC attrition. Liver (UBERON:0002107) is the primary site of TPO synthesis and is therefore the source organ of the deficient signal (though the liver itself is structurally normal). - **Secondary / complications:** Any tissue susceptible to hemorrhage from thrombocytopenia (skin/mucosa; CNS in severe cases). In biallelic marrow failure, generalized effects of anemia and neutropenia. - **Body system:** Hematopoietic/hematologic system (and, by extension, hemostatic system).

Tissue and cell level: - Hematopoietic tissue; specifically the megakaryocyte lineage (CL:0000556), hematopoietic stem cells (CL:0000037), and the end-product platelets (CL:0000233). All three hematopoietic lineages are affected in the biallelic marrow-failure state.

Subcellular level: - Plasma membrane receptor signaling platform (MPL receptor; GO:0005886 plasma membrane); cytoplasmic JAK-STAT signaling; the secretory pathway for TPO in hepatocytes.

Localization / lateralization: Systemic/bilateral — a diffuse hematopoietic process, not a localized or lateralized lesion.

---

8. Temporal Development

- **Onset:** Childhood in biallelic homozygotes (the index siblings presented with aplastic anemia in childhood); carriers exhibit lifelong, usually incidentally detected, mild thrombocytopenia. Onset is **insidious/chronic** rather than acute.
- **Progression:** The biallelic form is **progressive** toward marrow failure, mirroring the progressive HSC loss documented for MPL-inactivating mutations. Carrier thrombocytopenia is **stable**.
- **Disease course:** Chronic, lifelong. Without definitive treatment (HSCT) the biallelic marrow-failure form carries the morbidity/mortality of aplastic anemia.
- **Remission patterns:** No spontaneous remission; treatment-induced improvement is possible with TPO-RA (ligand bypass) or cure with HSCT.

- **Critical periods:** Early recognition before irreversible HSC depletion is the key window for intervention, by analogy to the reversibility demonstrated when MPL expression is restored in murine models.

---

9. Inheritance and Population

Epidemiology: Ultra-rare; prevalence and incidence are not formally established (essentially a single reported family plus mechanistic extrapolation). No reliable per-100,000 figures exist.

Inheritance: Autosomal recessive for the full disease (biallelic). A clear **gene-dosage effect** is observed, such that heterozygous carriers manifest mild thrombocytopenia — a semi-dominant feature at the level of platelet count.

- **Penetrance:** Biallelic genotype appears highly penetrant for thrombocytopenia; carrier phenotype is mild and may be subclinical.
- **Expressivity:** Variable — from mild isolated thrombocytopenia to aplastic anemia depending on residual TPO activity.
- **Genetic anticipation:** Not applicable (not a repeat-expansion disorder).
- **Germline mosaicism:** Not reported.
- **Founder effects / consanguinity:** The index family was consanguineous and of Micronesian ancestry; consanguinity is the principal route to the biallelic state. No broad founder haplotype is established.
- **Carrier frequency:** Unknown; expected to be very low.

Population demographics: Reported in a Micronesian family; no established ethnic predilection beyond the role of consanguinity. No sex bias is expected for an autosomal-recessive hematopoietic gene. Age distribution skews to childhood presentation for the severe form.

---

10. Diagnostics

Clinical/laboratory tests: - **Complete blood count:** Thrombocytopenia (isolated in mild cases; with anemia ± neutropenia/pancytopenia in biallelic marrow failure). - **Bone marrow examination:** Reduced/hypoproductive megakaryocytes; hypocellular marrow in the aplastic

form. - **Serum/plasma TPO level — a key discriminating test.** In most hypoproductive marrow-failure thrombocytopenias, TPO is **elevated** because of reduced receptor-mediated clearance when platelet/megakaryocyte mass is low. In THC9, the defect is the **ligand itself**, so TPO should be **low/absent or inappropriately normal** — an inverted pattern relative to classic aplastic anemia or CAMT. (The inverted expectation in THC9 is a mechanistic inference.)

> "plasma TPO levels markedly elevated in patients with AA or ChemoT" — PMID: 11345828
>
> "TPO is produced in the liver at a constant rate and cleared by TPO receptors on platelets. TPO levels are inversely proportional to the rate of platelet production." — PMID: 23821332
>
> "these new criteria enable us to clearly differentiate ITP from aplastic anemia and other forms of hypoplastic thrombocytopenia" — PMID: 37957517

Genetic testing (definitive): - **Single-gene *THPO* testing**, inherited-thrombocytopenia / bone-marrow-failure **gene panels**, **whole-exome sequencing** (the modality by which THC9 was originally discovered), and **whole-genome sequencing** are all appropriate. WES/WGS are especially valuable given the broad differential of inherited thrombocytopenias (>50 genes). - CMA, karyotyping, FISH, mtDNA, and repeat-expansion testing are **not relevant** to this single-nucleotide-variant, single-gene disorder.

Clinical criteria / differential diagnosis: Distinguish from: - **CAMT (MPL-related)** — receptor defect; elevated TPO. - **Immune thrombocytopenia (ITP)** — destruction, normal/slightly ↑ TPO, elevated immature platelet fraction, normal marrow. - Other inherited thrombocytopenias (e.g., *GFI1B*-related PMID: 41766040, *NFE2*-related PMID: 42152485) — distinguished by marrow morphology, associated features (α-granule deficiency, dysmegakaryopoiesis), and genetics. - Acquired marrow failure (aplastic anemia, MDS PMID: 26500138).

Screening: Cascade genetic testing of at-risk relatives in known families; carrier detection shows mild thrombocytopenia.

---

11. Outcome / Prognosis

- **Survival/mortality:** No THC9-specific survival statistics exist. Prognosis tracks with severity: isolated mild thrombocytopenia carries a good prognosis, whereas the biallelic

aplastic-anemia form carries the substantial morbidity/mortality of marrow failure (bleeding, infection) unless treated with HSCT.

- **Morbidity:** Bleeding risk (thrombocytopenia), anemia-related fatigue, and infection risk (neutropenia) in the severe form; transfusion dependence.
- **Complications:** Hemorrhage; in marrow failure, infections and transfusion-related complications; potential clonal evolution risk inherent to inherited marrow-failure syndromes (not specifically quantified for THC9).
- **Recovery potential:** The HSC-maintenance defect may be reversible if the TPO/MPL signal is restored early, by analogy to murine models in which restoring Mpl expression replenished the HSC pool and corrected platelet production.
- **Prognostic factors:** Residual TPO activity (allele severity), genotype (biallelic vs carrier), age at presentation, and degree of marrow involvement.

---

12. Treatment

Central therapeutic logic: THC9 is a **ligand deficiency with an intact MPL receptor**, making receptor-level pharmacology mechanistically rational.

Pharmacotherapy / TPO-receptor agonists (TPO-RA): - **Romiplostim** (NCIT:C66919) — a peptibody carrying four TPO-mimetic peptides that directly activate MPL. - **Eltrombopag** (NCIT:C62408) — an oral small-molecule MPL agonist acting via a distinct (transmembrane) mechanism.

> "A second generation of TPO receptor agonists, romiplostim and eltrombopag, has been developed." — PMID: 23821332
>
> "TPO is produced in the liver at a constant rate and cleared by TPO receptors on platelets. TPO levels are inversely proportional to the rate of platelet production." — PMID: 23821332

TPO-RAs are already applied to rare inherited thrombocytopenias and to aplastic anemia, supporting their use in TPO-ligand deficiency (PMID: 38925157). Pharmacodynamically, platelet counts rise after a ~5-day delay and peak at ~12–14 days; part of the benefit is anti-apoptotic (AKT–Bcl-xL) support of megakaryocytes/platelets (PMID: 25132654). A caveat for THC9 is that a ligand-replacement strategy and a receptor-agonist strategy are not identical: in

disorders where the block is *downstream* of the receptor (e.g., NFE2-related maturation defect), TPO-RAs fail (PMID: 42152485) — reinforcing that TPO-RA rationality in THC9 depends specifically on the receptor and its signaling being intact.

Advanced/curative therapeutics: - **Allogeneic HSCT** (NCIT:C15431) — curative for the marrow-failure end of the spectrum (as established for MPL-related CAMT). - **Gene therapy (proof of concept):** Lentiviral re-expression of Mpl in Mpl-deficient mice replenished the HSC pool and corrected platelet production, with demonstrated reversibility of HSC loss.

> "restoration of Mpl expression from transcriptionally targeted vectors prevents lethal adverse reactions of ectopic Mpl expression, replenishes the HSC pool, restores stem cell properties, and corrects platelet production" — PMID: 21289307

For THC9 specifically, the defect is the *ligand*, so either receptor agonism (bypassing the missing ligand) or, conceptually, *THPO* gene/protein replacement would be rational; the Mpl-restoration data establish the reversibility principle for the axis.

Supportive care: Platelet transfusion for acute bleeding; red-cell transfusion and infection prophylaxis in marrow failure.

Treatment outcomes / adverse events: The principal TPO-RA safety concerns are **thrombosis** and rare, usually reversible, marrow **reticulin fibrosis**.

> "Thrombotic complications are also an area of concern and need further close follow-up." — PMID: 22872157

Real-world ITP data show thrombosis typically occurs in patients with additional thrombotic risk factors (PMID: 27793553).

Personalized medicine: Genotype-guided — confirming a *THPO* ligand defect (vs an *MPL* receptor defect or a downstream transcription-factor defect) supports TPO-RA use, because receptor agonists cannot rescue a non-functional receptor or a downstream maturation block but can bypass a missing ligand.

---

13. Prevention

- **Primary prevention:** Not applicable to a monogenic disorder beyond **genetic counseling** and reproductive planning. In consanguineous families, carrier screening and counseling reduce the recurrence risk of the biallelic state.
- **Secondary prevention:** Early genetic diagnosis (cascade testing of relatives) and monitoring of platelet counts to pre-empt bleeding complications; early recognition before irreversible HSC depletion.
- **Tertiary prevention:** Prevent bleeding (platelet support), infection (in neutropenia), and manage marrow failure (HSCT) to prevent complications.
- **Immunization / public health / environmental / prophylaxis:** Not applicable as causal measures; standard infection prophylaxis applies in marrow-failure patients.
- **Counseling:** Autosomal-recessive recurrence risk (25% for carrier×carrier); carriers should be informed of mild thrombocytopenia and reproductive implications.

14. Other Species / Natural Disease

- **Taxonomy / orthologs:** *THPO* is conserved across mammals and in zebrafish. Mouse *Thpo* (NCBI Gene 21832) and its receptor *Mpl* are the principal experimental orthologs; zebrafish possess a thrombopoietin receptor (zTpoR) with partly divergent signaling.
- **Natural disease in other species:** No well-characterized spontaneous *THPO*-deficiency disease is cataloged in companion animals (OMIA did not surface a defined natural analog); the knowledge base is dominated by engineered models.
- **Comparative biology:** The TPO–MPL–JAK2 axis is evolutionarily conserved as the dominant thrombopoietic regulator, validating cross-species modeling. Zebrafish thrombopoiesis additionally engages an unconventional zTpoR–Mapk14a (p38) interaction not seen in human TpoR (PMID: 40089183), a caveat for cross-species extrapolation.
- **Transmission / zoonosis:** Not applicable (genetic disease).

15. Model Organisms

Mouse (primary model): - ***Thpo*-knockout and *Mpl*-knockout mice** recapitulate the core phenotype: profound thrombocytopenia while retaining ~10% of normal platelet counts of fully functional platelets, which prevents abnormal bleeding — demonstrating TPO as the dominant but not sole regulator of platelet production.

> "Even though these mice are profoundly thrombocytopenic they maintain a basal level of approximately 10% of the normal count of fully functional platelets." — PMID: 11012175

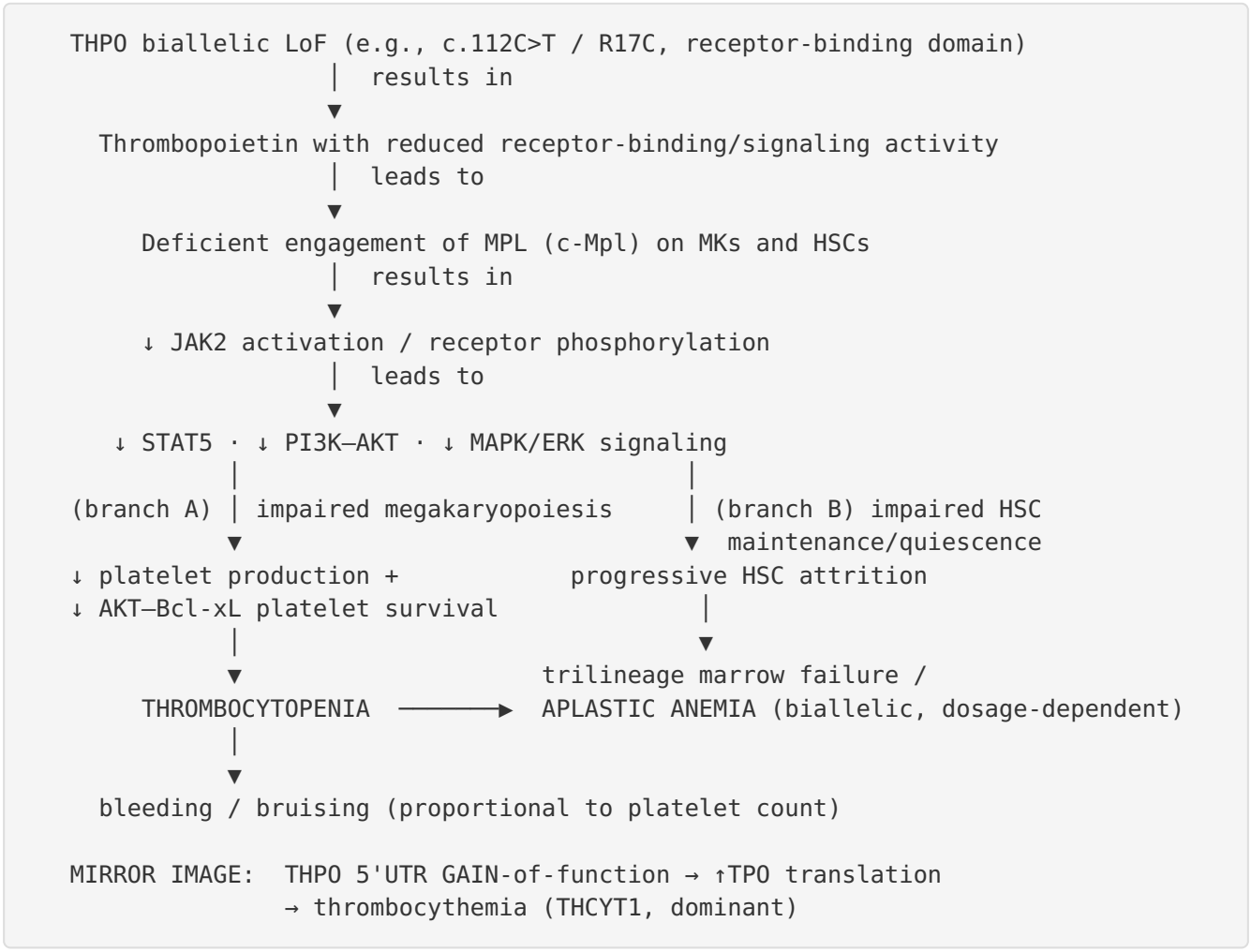
- IL-3Rα/Mpl double knockouts show no further platelet reduction, indicating the residual TPO-independent thrombopoiesis is not IL-3-driven (PMID: 11012175).
- *Mpl*-knockout mice also develop **aplastic anemia from progressive HSC loss**, which is **reversible** upon Mpl re-expression — modeling the marrow-failure extension of THC9 and establishing gene-therapy proof of concept (PMID: 21289307).
- Mpl cytoplasmic-tyrosine knock-in mutants (e.g., Y599/Y582) dissect which signaling outputs drive megakaryopoiesis vs HSC maintenance (PMID: 38491305, PMID: 32417303).

Phenotype recapitulation: High — murine TPO/MPL loss reproduces hypoproductive thrombocytopenia and the HSC/aplastic-anemia phenotype. **Limitations:** residual ~10% platelet production differs from the variable human severity; murine MPL responds less efficiently to human-designed TPO-RA (romiplostim), complicating preclinical drug testing (PMID: 32417303).

Other systems: Zebrafish thrombopoiesis (with the divergent zTpoR–Mapk14a axis) and in vitro TPO-dependent cell lines (UT7-TPO, used to quantify the R17C functional deficit; 32D cells for signaling). iPSC/organoid megakaryopoiesis systems are applicable but not yet reported specifically for THC9.

Resources: MGI (mouse *Thpo*/*Mpl*), ZFIN (zebrafish), Alliance of Genome Resources.

Mechanistic Model / Interpretation



The unifying concept is that **THPO gene dosage sets whole-body platelet mass as a bidirectional rheostat.** Coding loss-of-function turns the rheostat down (THC9: thrombocytopenia → aplastic anemia, recessive), while 5′UTR gain-of-function turns it up (hereditary thrombocythemia, dominant). Within THC9, a continuous dose–response from residual TPO activity (allele severity) to clinical severity explains why heterozygous carriers are mildly thrombocytopenic and biallelic homozygotes develop marrow failure. The intactness of the MPL receptor is the single most therapeutically consequential feature, because it means the downstream machinery is fully competent and only the input signal is missing — exactly the situation that receptor agonists (romiplostim, eltrombopag) and ligand/gene replacement are designed to correct.

Evidence Base

PMID	Title (abbrev.)	Role in this report
24085763	*Exome sequencing reveals a thrombopoietin ligand mutation in a Micronesian family with AR aplastic anemia*	**Founding evidence** — identifies THPO c.112C>T (R17C), partial LoF, dosage-dependent spectrum
38491305	*Differential in vivo roles of Mpl cytoplasmic tyrosines*	Defines TPO→MPL→JAK2 cascade; links to thrombocytopenia & HSC defects
21289307	*Lentiviral gene transfer regenerates HSCs in Mpl-deficient aplastic anemia*	HSC-loss mechanism; reversibility / gene-therapy proof of concept
11012175	*IL-3 does not contribute to platelet production in c-Mpl-deficient mice*	Knockout phenotype: ~10% residual platelets; TPO as dominant regulator
23821332	*The biology of thrombopoietin and TPO-RA*	TPO physiology/clearance; introduces romiplostim & eltrombopag
11345828	*Diagnostic value of TPO levels...*	Marrow-failure thrombocytopenias show ↑TPO (contrast to THC9 ↓TPO)
37957517	*Reference guide for ITP diagnosis, 2023*	TPO/IPF to differentiate hypoplastic thrombocytopenias
9694695	*Familial ET with 5'UTR THPO deletion*	Opposite GoF mechanism (thrombocythemia)
10828006	*Translational pathophysiology*	uAUG/5'UTR translational control of TPO
38925157	*TPO-RA in pediatrics*	TPO-RA use in rare inherited thrombocytopenias/SAA
22872157	*Thrombopoietin-receptor agonists*	Thrombosis as key TPO-RA adverse event
32417303	*Signaling of murine MPL after TPO/ romiplostim*	Preclinical caveat: murine MPL less responsive to romiplostim
25132654	*TPO-RA effect on platelet apoptosis*	Anti-apoptotic AKT–Bcl-xL contribution

Supporting / contextual (differential and adjacent biology): PMID: 41766040 (GFI1B-related IT), PMID: 42152485 (NFE2-related, TPO-RA-refractory — highlights a downstream defect distinct from THC9), PMID: 26500138 (MDS thrombocytopenia), PMID: 18496561 (familial ET / JAK2), PMID: 40089183 (zebrafish zTpoR–Mapk14a), PMID: 35665097 (TPO in sepsis — diagnostic confounder).

Limitations and Knowledge Gaps

1. **Narrow human evidence base.** The human characterization of THC9 rests essentially on a **single consanguineous family** with one missense allele (R17C). The full mutational spectrum (nonsense/frameshift/splice nulls), the range of phenotypes, penetrance, and natural history are therefore not well defined.
2. **Epidemiology unknown.** No prevalence/incidence estimates, no defined carrier frequency, no established ethnic distribution beyond the consanguinity route.
3. **Serum TPO in THC9 is an inference.** The expectation of low/inappropriately-normal TPO (vs the elevated TPO of other marrow-failure states) follows from the ligand-loss mechanism but has not been robustly measured in a THC9 cohort — and a receptor-binding-domain missense variant could still produce detectable (but non-functional) protein, complicating immunoassay interpretation.
4. **Treatment data are extrapolated.** No THC9-specific clinical trials of TPO-RA, HSCT, or gene therapy exist; efficacy is inferred from TPO/MPL biology and from related disorders (CAMT, SAA, ITP). TPO-RA response in a ligand defect with a missense (partial) mutation is plausible but unproven in patients.
5. **Model-organism caveats.** Murine MPL responds less efficiently to the human-designed agonist romiplostim (PMID: 32417303), and zebrafish signaling diverges (PMID: 40089183), limiting direct preclinical translation of drug responses.
6. **Genotype–phenotype quantitation.** The relationship between residual TPO activity and clinical severity is qualitatively clear (dosage effect) but not quantitatively mapped.

Proposed Follow-up Experiments / Actions

1. **Case ascertainment & registry.** Use GeneMatcher/Matchmaker Exchange and inherited-thrombocytopenia panels to find additional biallelic *THPO* families; build a small natural-history registry to define penetrance, severity distribution, and progression to marrow failure.
2. **Measure serum/plasma TPO in confirmed THC9 patients** (ideally with both immunoassay and a bioassay) to test the predicted low/non-functional-ligand pattern and establish TPO level as a discriminating diagnostic (vs elevated TPO in CAMT/aplastic anemia).
3. **Functional characterization of new alleles.** Express candidate *THPO* variants in UT7-TPO / 32D systems and in iPSC-derived megakaryopoiesis to quantify residual activity and correlate with phenotype.
4. **Therapeutic proof-of-concept.** Trial a TPO-RA (eltrombopag/romiplostim) in a confirmed THC9 patient with monitoring of platelet response, marrow reticulin, and thrombosis risk; compare the response of a ligand defect with the receptor-defect (CAMT) and downstream-defect (NFE2/GFI1B) comparators, which help define where TPO-RA will and will not work.
5. **Ligand/gene replacement modeling.** Test recombinant TPO or *THPO* gene replacement in a *Thpo*-null mouse and in patient iPSC-megakaryocyte systems to establish whether supplying ligand (rather than agonizing the receptor) better restores HSC maintenance.
6. **Diagnostic algorithm integration.** Incorporate "low/normal TPO + central thrombocytopenia + *THPO* biallelic variant" into inherited-thrombocytopenia diagnostic guidelines to distinguish THC9 from ITP, CAMT, and acquired marrow failure.

Evidence source types: human clinical genetics (index family, *24085763* *); model organism (mouse Thpo/Mpl knockouts and gene-therapy models; zebrafish); in vitro (UT7-TPO/32D functional and signaling assays); and mechanistic/clinical pharmacology reviews. Where claims rest on inference from TPO/MPL biology rather than direct THC9 observation, this is stated explicitly.*
